

Pose-Based Isolated Sign Recognition in the Browser: Signer-Independent Evaluation and Cross-Lingual Transfer from ASL to French Belgian Sign Language

Anais Azouaoui

Abstract

We study pose-based isolated sign recognition under the condition that matters for deployment: recognising signs produced by people never seen during training. From MediaPipe Holistic landmarks, we build a single, unit-tested preprocessing function shared by training and by a JavaScript port that runs a real-time demo entirely in the browser. On the Google ISLR dataset (250 American Sign Language signs, 21 signers), a 2.2 M-parameter convolution–Transformer reaches $64.8 \pm 0.4\%$ top-1 accuracy on four unseen signers, against $63.0 \pm 0.5\%$ for a bidirectional GRU and 42.1% for a statistical baseline, while a random split would overestimate the accuracy by 16.9 points. Ablations show that mirroring every clip to a canonical dominant hand matters most, whereas lips, longer sequences, depth and a three times longer training do not help, and that early stopping can inflate the apparent effect of an ablation fivefold. On LSFB-ISOL (100 glosses of French Belgian Sign Language cut from conversations), we find that initialising the model from ASL brings no benefit at any training-set size, from 5 examples per sign to the full corpus, where training from scratch is better (68.3% vs. 66.2% on 40 unseen signers). We also report three pitfalls that silently degrade such pipelines: augmentations cancelled by normalisation, anisotropic coordinates of anamorphic videos, and feature outliers that break INT8 quantisation, which moreover makes the model slower on CPU. The browser demo classifies a sign in 16 ms on one CPU thread and reproduces the PyTorch predictions on every tested clip.

1 Introduction

Sign languages are complete natural languages, with their own lexicon and grammar, used by millions of Deaf and hard-of-hearing people. Automatic sign language processing covers a spectrum of tasks of increasing difficulty: *isolated* sign recognition (one sign per clip, a closed vocabulary), continuous recognition, and translation to and from spoken languages [5, 16, 28]. This work addresses the first of these tasks only. We stress this point because isolated recognition is regularly presented as “sign language translation”: a classifier over a few hundred glosses ignores the grammar of the language, its use of space and its non-manual markers, and cannot translate anything [28].

Within isolated sign recognition (ISLR), *pose-based* approaches first extract body, hand and face landmarks with an off-the-shelf estimator such as MediaPipe Holistic [2, 15, 19, 30], then classify the resulting sequence of keypoints. Compared with RGB models, they are small, fast, largely insensitive to background and appearance, and they let a system work without transmitting video, an important property for users’ privacy. The Google *Isolated Sign Language Recognition* competition [8] popularised this setting with 94,477 landmark sequences of 250 American Sign Language (ASL) signs.

Three gaps motivate this study. **(i) Evaluation.** Sign recognition models must generalise to people they have never seen, but random splits mix the clips of each signer between training and test, which inflates the scores. We evaluate every model on held-out signers and quantify the gap with a random split. **(ii) Low-resource sign languages.** Most resources concern ASL, whereas French Sign Language (LSF) and French Belgian Sign Language (LSFB) have far less labelled data. We ask how much a model pre-trained on ASL helps on LSFB-ISOL [11], a corpus of signs cut from spontaneous conversations, especially when only a handful of examples per sign are available. **(iii) Deployment.** A recognition model

is only useful if it runs where users are. We target a web browser on a laptop without GPU, which requires the preprocessing used during training to be reproduced exactly in JavaScript.

Contributions.

- A reproducible pipeline built around a single, unit-tested preprocessing function shared by training, evaluation and a JavaScript port, whose outputs match the Python reference to within 3×10^{-6} (relative error) on synthetic test cases (Section 4.5).
- A signer-independent comparison of a statistical baseline, a bidirectional GRU and a convolution–Transformer hybrid on 250 ASL signs, with three seeds, a quantification of the leakage caused by random splits and one-factor ablations of every preprocessing choice (Sections 6.1 and 6.2).
- A controlled study of cross-lingual transfer from ASL to LSFB, with learning curves from 5 to all available examples per sign. Contrary to our expectation, initialising from ASL brings no benefit at any data size and costs 2.0 points with all data (Section 6.3).
- A real-time, fully client-side demo in which the Conv-Transformer classifies a sign in 16 ms on one CPU thread of a laptop browser, with automatic sign segmentation; and the observation that dynamic INT8 quantisation, while making the model 3.3× smaller, first destroys its accuracy because of rare feature outliers and then remains 2.1× slower on CPU (Section 6.4).

Code, configuration files and the frozen evaluation splits are released with the paper.

2 Related work

Isolated sign recognition datasets. Large ASL vocabularies are available in WLASL [17] (2,000 glosses collected from the web), MS-ASL [25] and ASL Citizen [10], the latter being community-sourced with explicit consent and signer-independent splits. BSL-1K [1] mines British Sign Language signs from broadcast footage using mouthing cues. The Google ISLR competition [8] released MediaPipe landmarks only, recorded by 21 participants on smartphones, which removes the need for video storage. For French-related sign languages, LSFB-ISOL and LSFB-CONT [11] are derived from the LSFB corpus [20], and MEDI-API-SKEL [6] and Dicta-Sign-LSF-v2 [3] provide continuous LSF with skeletons or annotations, but no isolated-sign benchmark.

Pose-based models. Skeleton sequences have been classified with spatio-temporal graph convolutions [14, 24, 27], with Transformers operating directly on keypoints [4] and with self-supervised pre-training of hand representations [13]. The strongest solutions of the Google ISLR competition combined depthwise temporal convolutions with a few self-attention layers, close in spirit to the Conformer used in speech recognition [12]; our Conv-Transformer follows this design at a smaller scale. Recurrent networks [7] remain a light and strong baseline.

Transfer across sign languages. Sign languages share many articulatory primitives (handshapes, movements, locations), which suggests that features learned on a large corpus may transfer to a smaller one, in the sense of Yosinski et al. [29]. Large-scale pre-training improves recognition on smaller datasets of the same language [1]; cross-lingual transfer between unrelated sign languages from pose data alone has received less attention, and we provide a controlled study with learning curves.

Responsible deployment. We follow the recommendations of Bragg et al. [5] and Yin et al. [28]: we describe the system as isolated sign recognition, document its limits in a model card [21] and process video locally on the user’s device.

Table 1. Datasets after conversion to the unified format. Durations are medians; “missing hands” is the share of frames in which neither hand is detected.

Dataset	Classes	Clips	Signers	Signers tr/val/test	Clips per class	Duration	fps	No hand
ASL (Google ISLR)	250	94,477	21	14/3/4	299–415	0.73 s	30	34 %
LSFB-ISOL	100	55,286	99	50/9/40	216–3293	0.32 s	50	2 %

3 Data

Google ISLR (ASL). The competition data [8] contain 94,477 sequences of 250 ASL signs produced by 21 participants, with the 543 MediaPipe Holistic landmarks (468 face, 33 body, 21 per hand) of every frame and no video. Signs were recorded with a smartphone in a dictionary-like setting, one sign per clip. The landmarks were produced by the legacy MediaPipe Holistic *solution*, an older pipeline than the MediaPipe Tasks API we use in the browser (Section 4.5).

LSFB-ISOL (French Belgian Sign Language). LSFB-ISOL v2 [11] contains 120,739 isolated signs cut from spontaneous conversations of the LSFB corpus [20], covering 4,657 glosses and 99 signers recorded at 50 fps. Landmarks are provided per body part, extracted with the MediaPipe Tasks models (33 body and 478 face points). Because the glosses follow a long-tailed distribution, we keep the 100 most frequent glosses performed by at least ten different signers. We use the *raw* landmarks, without the interpolation and smoothing applied to the default release, so that missing detections appear as they would during live use; clips with fewer than four frames (0.6 % of the selected clips, most of them empty) are discarded. LSFB is related to LSF but is a distinct language: we never present LSFB results as LSF results.

Unified landmark subset. Both datasets are converted to a common representation of $L = 89$ landmarks per frame: the two hands (2×21), seven upper-body points (nose, shoulders, elbows, wrists) and 40 points of the inner and outer lip contours, which carry part of the mouthing and non-manual information. Hips are excluded because they are usually outside the camera field in both datasets. A sequence is a tensor $X \in \mathbb{R}^{T \times L \times 3}$ of normalised image coordinates (x, y, z) with NaN entries for undetected landmarks. Converting the raw ASL data (more than 40 GB of parquet files) to this format is done once in a cloud notebook and yields a 2 GB file; LSFB poses are streamed from the dataset server and reduced on the fly.

Aspect ratio. MediaPipe normalises x by the frame width and y by the frame height, so distances are only comparable across sources once y is multiplied by the height-to-width ratio of the displayed image. The LSFB videos are 720×576 PAL frames with a 16:9 display aspect, which inflates vertical distances by $1024/576 \approx 1.78$. We checked this on the data with an anatomical invariant, the height of the nose above the shoulder line divided by the shoulder span: its median is 0.40 in ASL and 0.77 in raw LSFB, but 0.43 after correction (a 4:3 assumption would give 0.57). We therefore scale y by 0.5625 for LSFB, by 1 for ASL, whose coordinates are close to isotropic, and by the exact frame ratio for webcam and video input in the demo. Figure 4 shows that, after this correction and the normalisation of Section 4.1, both corpora share the same body frame.

Figure 1 contrasts the two corpora. ASL classes are balanced and clips are recorded in isolation, whereas LSFB classes span more than an order of magnitude and clips are short (median 0.32 s against 0.73 s), because conversational signs are co-articulated with their neighbours. Missing hands are far more frequent in ASL (Figure 2): one of the two hands is absent from most frames, and neither is detected in about a third of them. These are not idle frames: in a random sample of 5,000 clips, only 2.4 % of the hand-less frames lie before or after the sign; the others fall in the middle of it, when fast motion or the narrow smartphone framing makes the hand detector fail. Figure 3 shows one clip of each dataset.

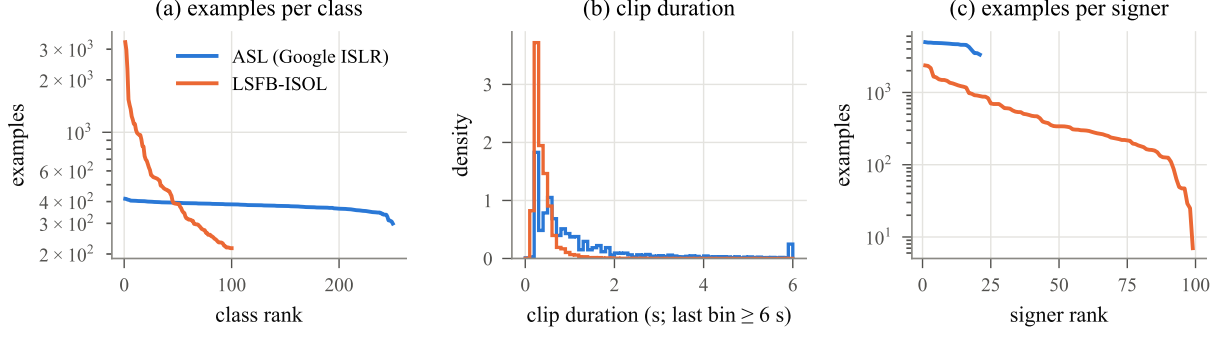


Figure 1. Dataset overview. (a) Examples per class, sorted; (b) distribution of clip durations; (c) examples per signer, sorted.

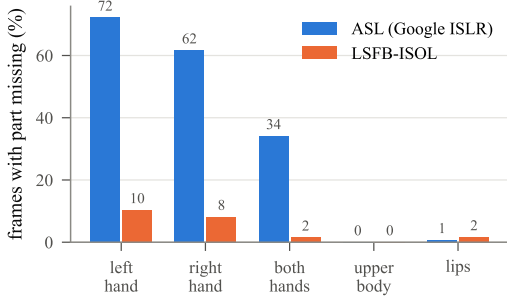


Figure 2. Share of frames in which a body part is not detected.

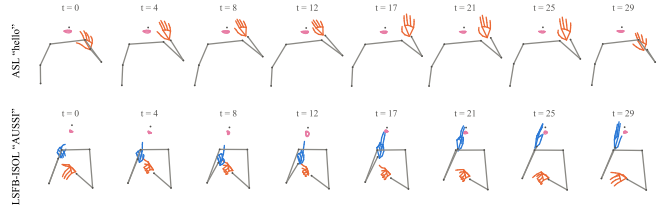


Figure 3. One ASL and one LSFB clip, eight evenly spaced frames (hand slots in orange and blue, lips in pink, upper body in grey).

4 Method

Figure 5 summarises the system. A sequence of raw landmarks is transformed by a deterministic preprocessing function into a fixed-size feature matrix, augmented at training time only, and classified by a sequence model.

4.1 Preprocessing

Let $X \in \mathbb{R}^{T \times L \times 3}$ be a raw sequence. The preprocessing applies, in order:

Temporal trimming. Leading and trailing frames in which no hand is detected are removed (the sequence is kept as is if no hand is ever detected).

Canonical dominant hand. Signers use their dominant hand for one-handed signs, and the handedness labels of the landmark estimator depend on whether the camera image is mirrored. We count the frames in which each hand slot is detected and, if the left slot is the more frequent, mirror the sequence: $x \mapsto 1 - x$ for all points, together with the permutation π that swaps left and right hands, shoulders, elbows, wrists and symmetric lip points. After this step the features are invariant to a horizontal flip of the input video whenever one hand dominates, which we verify by a unit test.

Spatial normalisation. Let s_t^l, s_t^r be the shoulder positions in frame t and $\mathcal{T}_s$ the frames in which both are detected. We centre and scale all points by

$$c = \frac{1}{|\mathcal{T}_s|} \sum_{t \in \mathcal{T}_s} \frac{s_t^l + s_t^r}{2}, \quad w = \frac{1}{|\mathcal{T}_s|} \sum_{t \in \mathcal{T}_s} \|s_t^l - s_t^r\|_2, \quad \tilde{X}_{t,j} = \frac{X_{t,j} - c}{w}, \quad (1)$$

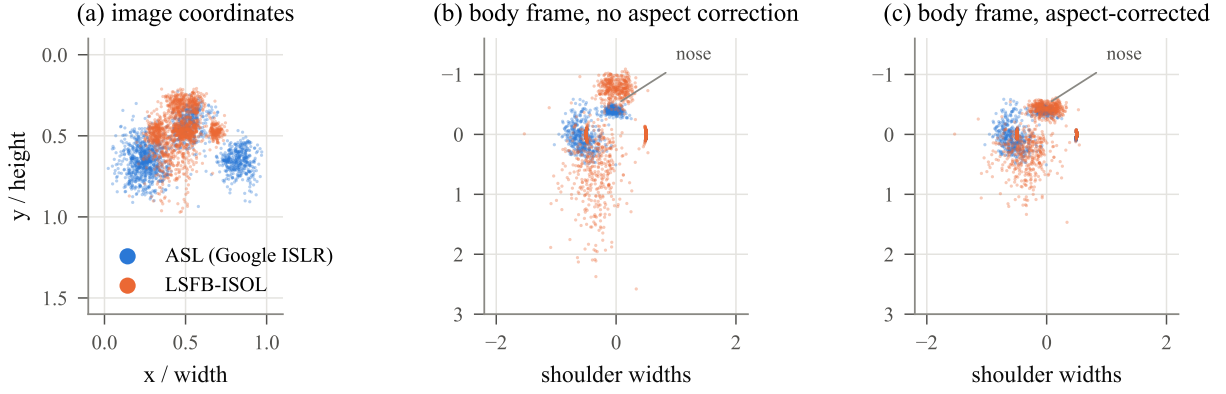


Figure 4. Per-clip mean position of the nose, the shoulders and the dominant wrist for 400 random clips of each corpus. (a) Image coordinates; (b) body frame without aspect correction: the LSFB nose sits twice as high as the ASL one; (c) body frame with the aspect correction: both corpora align.

in the browser: *MediaPipe Tasks JS + preprocess.js + ONNX Runtime Web*

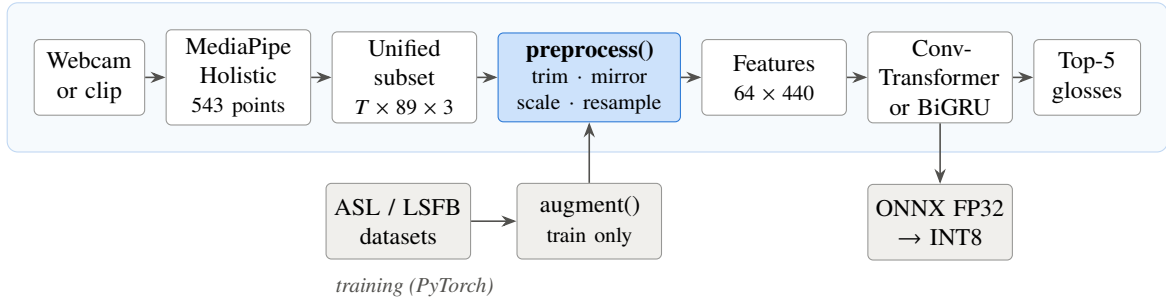


Figure 5. Overview. The same preprocessing function feeds training (Python) and the browser runtime (JavaScript port, checked for parity). Only landmarks are processed; video never leaves the device.

applied to (x, y) , with z divided by w . The constants are computed over the whole clip, so that the global motion of the hands relative to the body is preserved. If the shoulders are missing ($w < 0.02$), we fall back to the centroid of all visible points and four times their average standard deviation.

Fixed length. The clip is linearly resampled to $T' = 64$ frames. Resampling also normalises the signing speed and the frame rate (30 fps for ASL, 50 fps for LSFB). A value whose interpolation involves a missing neighbour takes the value of the nearest frame. We compare this choice with zero-padding and masking in Section 6.2.

Features. Each frame is described by (i) the normalised (x, y) coordinates of the 89 landmarks; (ii) the dominant hand expressed relative to its wrist and divided by the length of the wrist–middle-finger-base segment, which isolates the handshape from the hand position; and (iii) the first-order temporal differences of (i) and (ii). The depth coordinate is dropped by default. This gives $F = 2 \times (89 + 21) \times 2 = 440$ features per frame. Missing values are finally replaced by zeros; the model sees missing landmarks as exact zeros at the body centre.

4.2 Augmentation

Augmentations are applied at training time only, in two stages. On the raw sequence we apply a time warp, which combines a global speed change ($\sigma \sim \mathcal{U}[0.8, 1.2]$, the clip is resampled to T/σ frames) with a smooth monotonic warp of the time axis made of four segments whose relative speeds are drawn in $[0.7, 1.3]$; random frame dropping ($p = 0.1$); random masking of individual landmarks ($p = 0.05$) and of a whole hand in a frame ($p = 0.05$); and Gaussian noise ($\sigma = 0.002$). The non-uniform part matters:

since every clip is resampled to T' frames, a uniform speed change alone would be almost entirely undone by the length normalisation.

With probability 0.8, a spatial transform is then applied *after* the shoulder normalisation, in the body frame: a rotation of $\pm 15^\circ$, per-axis scaling in $[0.8, 1.2]$ and a translation of up to 0.1 shoulder widths. Applying it before the normalisation, in image coordinates, would be pointless for translations and isotropic scaling, which the normalisation cancels exactly (a property checked by our unit tests). Augmentations are seeded by (seed, epoch, sample) for reproducibility.

4.3 Models

Input projection. Both sequence models first map each frame to $d = 192$ dimensions with two linear layers, layer normalisation and GELU activations.

BiGRU. Two bidirectional GRU layers [7] with 192 hidden units per direction, followed by the concatenation of masked mean and max pooling over time and a linear classifier (1.43 M parameters).

Conv-Transformer. Two stages, each made of three convolution blocks followed by one Transformer block [26], after a learned positional embedding. A convolution block is a pre-norm residual unit: pointwise expansion with a gated linear unit [9], a depthwise temporal convolution of kernel 17, batch normalisation, SiLU and a pointwise projection, similar to the convolution module of the Conformer [12]. The Transformer blocks use four heads and an MLP ratio of two. The sequence is mean-pooled under the mask before the classifier (2.16 M parameters). Attention is implemented with plain matrix products so that the model exports to ONNX without custom operators.

4.4 Training

Models are trained with AdamW [18] (learning rate 10^{-3} , weight decay 0.05, batch size 128), a cosine schedule with three warm-up epochs, label smoothing of 0.1 [23] and gradient clipping at norm 1, for at most 50 epochs with early stopping on the validation macro-F1 (patience 10). We keep the checkpoint with the best validation macro-F1. Mixed precision is used on GPU.

Transfer. For cross-lingual transfer, all weights except the classifier are initialised from the ASL model; a new classifier is trained for the target vocabulary. We compare full fine-tuning with a variant that freezes the encoder for the first five epochs and, as robustness checks, with fine-tuning at a five times lower learning rate and with a linear probe (encoder frozen during the whole training). Because the input representation is identical across datasets, no adapter is needed.

4.5 Browser runtime

The demo runs the MediaPipe Holistic Landmarker (Tasks API, WebAssembly with GPU delegate) on the webcam stream, builds the 89-landmark frame, and feeds a segmenter that detects the start and end of a sign. The segmenter is a small state machine: a sign starts after three consecutive frames with a visible hand, and ends when the hands disappear for six frames, stay still (mean hand displacement below 0.012 shoulder widths per frame) for fourteen frames, or after 120 frames. The segment is then preprocessed by a JavaScript port of the Python function and classified by the ONNX model with ONNX Runtime Web [22] (WebAssembly backend).

Discrepancies between training and deployment preprocessing are a frequent cause of failing demos. We therefore generate test fixtures from the Python implementation (missing hands, a mirrored input, missing shoulders, single-frame and long clips, padding, depth features) and check in the test suite that the JavaScript port reproduces the features to within 3×10^{-6} , the precision of single-precision arithmetic. We also check the complete browser path end to end: run in headless Chromium through the demo’s own

Table 2. ASL results (250 signs). The validation and test sets contain 3 and 4 signers never seen during training. Sequence models: mean $\pm$ standard deviation over three seeds, 50 training epochs; the last column is the standard deviation of top-1 accuracy across the four test signers. All values in %.

Model	Params	Validation	Test (4 unseen signers)			
		Top-1	Top-1	Top-5	Macro-F1	Signer s.d.
Chance	–	0.40	0.40	2.0	–	–
Statistics + log. regression	–	47.9	42.1	69.4	41.9	–
Statistics + LightGBM	–	45.2	33.8	61.5	33.4	–
BiGRU	1.43 M	65.8 $\pm$ 0.5	63.0 $\pm$ 0.5	84.0 $\pm$ 0.3	62.3 $\pm$ 0.6	3.9 $\pm$ 0.4
Conv-Transformer	2.16 M	68.2 $\pm$ 0.1	64.8 $\pm$ 0.4	84.0 $\pm$ 0.3	64.1 $\pm$ 0.4	3.3 $\pm$ 0.2

code, the LSFB test clips shipped with the demo receive the same top-1 prediction as with the PyTorch checkpoint (40/40 clips), with probabilities that differ by at most 2×10^{-6} .

5 Experimental protocol

Splits. All splits are *signer-independent* and frozen in version-controlled files. For ASL, the 21 participants are divided into 14 training, 3 validation and 4 test signers. For LSFB we keep the official test signers and draw 15 % of the official training signers for validation (Table 1). To measure leakage, we also build a random, sample-level ASL split with the same proportions, in which every signer appears in all subsets.

Metrics. We report top-1 and top-5 accuracy, macro-F1 over classes, and the mean and standard deviation of top-1 accuracy across test signers. Model selection and all ablations use the validation signers; the test signers are used once, for the final models.

Seeds and compute. Main models are trained with three seeds and reported as mean $\pm$ standard deviation; ablations use one seed. Deep models are trained on a single NVIDIA T4 GPU of a cloud notebook, except three transfer checks run on the laptop GPU (Apple M1) once the cloud quota was exhausted; latency is measured on the laptop CPU with one thread.

Baselines. Chance level; and per-coordinate temporal statistics (mean, standard deviation, minimum and maximum of the normalised position features over the clip, 4×220 features) classified by multinomial logistic regression or by gradient-boosted trees (LightGBM).

6 Results

6.1 ASL: baselines and sequence models

Table 2 reports the main comparison. Temporal statistics already carry much information: a logistic regression reaches 42.1 % top-1 on unseen signers, a hundred times chance level. Gradient-boosted trees fit the training signers better but generalise worse (33.8 % on test, 11 points below their validation score, against 6 points for the linear model): with only 14 training signers, flexible models on hand-crafted statistics latch onto signer-specific cues.

Both sequence models outperform the baselines by more than 20 points. The Conv-Transformer reaches 64.8 ± 0.4 % top-1 on the test signers, against 63.0 ± 0.5 % for the BiGRU, with an identical top-5 accuracy of 84 %. Seeds have little influence (standard deviations below one point), whereas test signers do: accuracy ranges from 61.9 to 69.5 % across the four of them, and the Conv-Transformer is better than the BiGRU for each one (Figure 7a). The validation accuracy keeps rising until the end of the 50-epoch

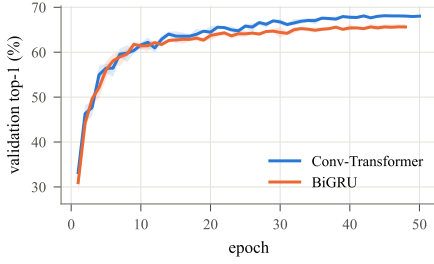


Figure 6. Validation top-1 during training (mean over seeds, band: min–max).

Table 3. Most frequent confusions of the Conv-Transformer on the ASL test signers (seed 0). Rate: share of the clips of the true sign.

True sign	Predicted	Count	Rate
wake	awake	40	51 %
pen	pencil	36	45 %
think	penny	33	43 %
awake	wake	31	42 %
duck	goose	26	32 %
look	see	24	26 %
say	chin	23	32 %
scissors	cut	22	33 %
have	animal	22	31 %
listen	hear	21	22 %
lips	mouth	21	28 %
there	boat	20	27 %

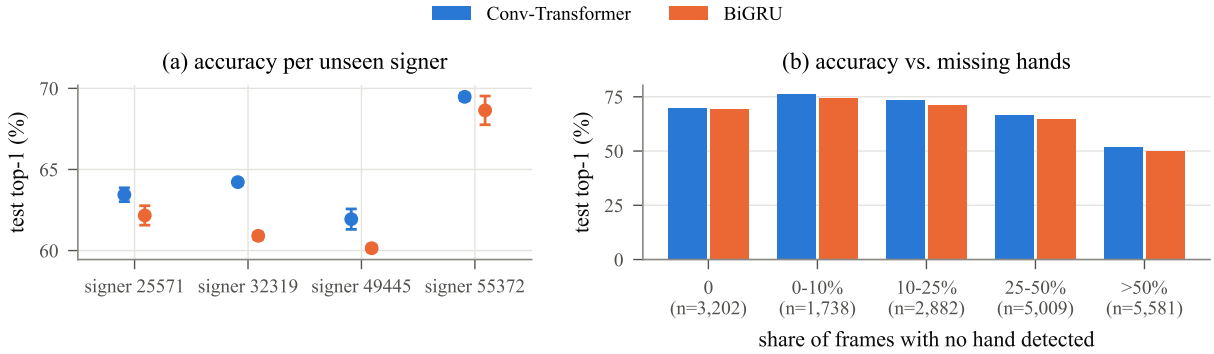


Figure 7. (a) Test top-1 for each unseen signer (mean and standard deviation over seeds). (b) Test top-1 as a function of the share of frames in which neither hand is detected.

schedule (Figure 6), but a three times longer schedule does not improve it (Section 6.2): the late gains come from the decay of the learning rate.

Error analysis. The most frequent confusions (Table 3) are linguistically meaningful. Several pairs are near-synonyms that share the same or a very similar sign in many ASL varieties (WAKE/AWAKE, PEN/PENCIL, LOOK/SEE), and others differ by a small change of handshape or location (DUCK/GOOSE, THINK/PENNY). The hardest classes are THERE (3 %), FINGER (12 %), GO (18 %): pointing and directional signs whose form depends on where the referent is, which a closed-vocabulary classifier cannot resolve. Part of the residual error is therefore a property of the vocabulary rather than of the model.

The quality of the landmarks is the other main factor (Figure 7b). Accuracy reaches 76.5 % when hands are missing in at most 10 % of the frames, but falls to 52.0 % when no hand is detected in more than half of the frames, a situation that concerns 30 % of the test clips. Clips in which a hand is visible in every frame are slightly less accurate than the next bin because they are shorter (median 16 frames instead of 24): among them, clips of at most ten frames, where the sign is probably truncated, are recognised in only 58 % of the cases, against 77 % for clips of 20 to 40 frames.

6.2 Leakage, ablations and training budget

Leakage. Evaluating on held-out signers is not a detail. With a random, sample-level split of the same size, in which every signer appears in training and test, the Conv-Transformer reaches 81.7 % test top-1 instead of 64.8 %: a random split overestimates the accuracy on new users by 16.9 points, a larger effect

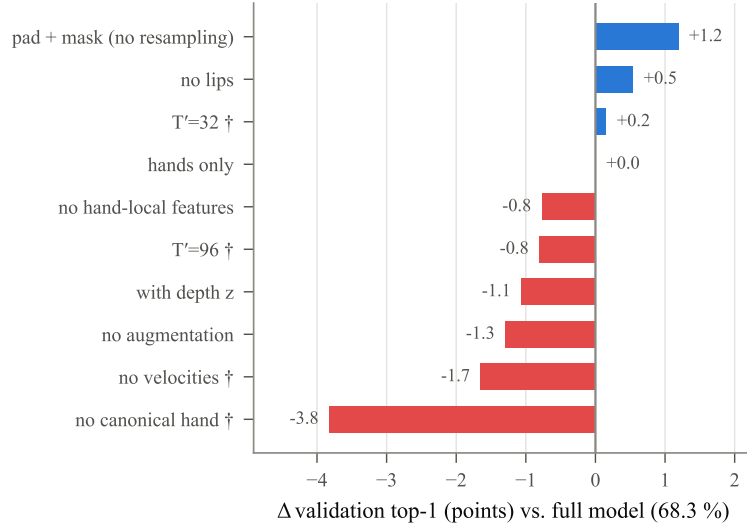


Figure 8. One-factor ablations of the Conv-Transformer on the ASL validation signers (seed 0). †: early stopping ended the run before the end of the 50-epoch schedule (epochs 41–46), so the effect may be overestimated.

than any architectural or preprocessing choice studied here.

Ablations. Figure 8 changes one factor at a time with respect to the full model (68.3 % validation top-1; seed-to-seed standard deviation 0.1 point). Early stopping turned out to be a confounder. Two variants first stopped on a validation plateau after 21 and 31 epochs, while the learning rate was still high, and seemed to lose 4.4 and 2.7 points; re-run for the full 50-epoch schedule, their effects shrink to -0.8 (no hand-local features) and -1.1 (depth added) points. The other variants marked with † (4 in total) stopped between epochs 41 and 46; their effects may be slightly overestimated, and we could not re-run them within our compute budget.

With this caveat, the largest effect is the canonical dominant hand (-3.8 points without it): otherwise, the same sign appears in either hand slot depending on the signer’s handedness and on mirroring. Velocities (-1.7), augmentation (-1.3), depth (-1.1, consistent with monocular depth estimates being noisy) and the hand-local shape features (-0.8) have moderate effects.

Three results contradict common practice. First, the lips and the upper body bring no measurable gain on this ASL vocabulary (+0.5 without lips, +0.0 with hands only): the 250 signs are mostly distinguished by the hands, and mouthing is presumably rare in dictionary-like recordings. Second, shorter sequences are as good ($T' = 32$: +0.2) and longer ones not better ($T' = 96$: -0.8), the median clip having only 22 frames. Third, zero-padding with a mask instead of resampling improves accuracy (+1.2): resampling stretches short clips and removes the duration cue, whereas padding keeps the true temporal dynamics. We keep the resampled configuration in the rest of the paper because it was fixed before the ablations, but padding is the better choice for future work.

Longer training and ensembles. Training the Conv-Transformer for 150 instead of 50 epochs does not help (67.5 % against 68.3 % on validation): the late rise of the training curves in Figure 6 comes from the annealing of the learning rate, not from under-training. Averaging the predictions of the three seeds, on the other hand, is a free improvement: 67.4 % test top-1 on ASL (64.8 % for a single model) and 70.9 % on LSFB (68.3 %), at three times the inference cost.

6.3 Cross-lingual transfer to LSFB

Table 4 reports the results on LSFB-ISOL with all training data. The Conv-Transformer trained from scratch reaches 68.3 % top-1 and 63.9 % macro-F1 on the 40 official test signers, far above the statistical

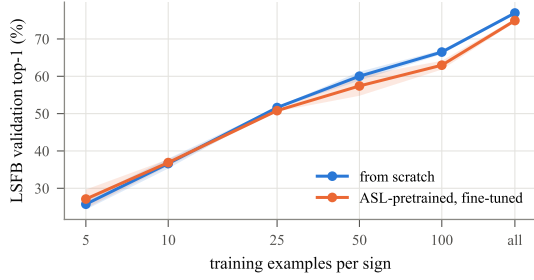


Figure 9. LSF validation top-1 as a function of the number of training examples per sign (mean $\pm$ s.d. over three seeds). “All” corresponds to a median of 185 examples per sign.

Table 4. LSF-ISOL, 100 glosses, all training data. Mean $\pm$ s.d. over three seeds, values in %.

Model	Validation	Test (40 unseen signers)		
	Top-1	Top-1	Top-5	Macro-F1
Chance	1.0	1.0	5.0	–
Statistics + log. regression	57.0	49.0	75.3	40.9
Statistics + LightGBM	54.8	47.1	72.6	37.3
Conv-Transformer, from scratch	77.0 $\pm$ 0.3	68.3 $\pm$ 0.2	87.7 $\pm$ 0.1	63.9 $\pm$ 0.2
ASL-pretrained, fine-tuned	74.9 $\pm$ 0.4	66.2 $\pm$ 0.2	87.1 $\pm$ 0.4	61.7 $\pm$ 0.3
ASL-pretrained, encoder frozen 5 epochs	75.6 $\pm$ 0.6	66.6 $\pm$ 0.2	86.5 $\pm$ 0.5	62.0 $\pm$ 0.3

Table 5. Model size and single-sign latency on an Apple M1 CPU, one thread, after export to ONNX. Native: ONNX Runtime; browser: ONNX Runtime Web (WebAssembly) in Chromium. INT8: dynamic quantisation of the weight matrices, the input projection being kept in FP32. Validation top-1 on 2,000 ASL validation clips.

Model	Precision	Size	Native	Browser (WebAssembly)		Val. top-1
			p50 (ms)	p50 (ms)	p95 (ms)	
Conv-Transformer	FP32	8.3 MB	7.8	16.3	22.0	67.8
Conv-Transformer	INT8	2.5 MB	16.5	33.4	44.8	68.3
BiGRU	FP32	5.5 MB	5.7	7.9	11.0	–
BiGRU	INT8	4.8 MB	5.5	8.1	10.2	–

baselines (49.0 % and 40.9 %). Macro-F1 is markedly lower than top-1 on this corpus because of its long tail: the ten most frequent glosses make up 32 % of the test clips.

Initialising the encoder from the ASL model does not help. With all the data, fine-tuning the ASL model reaches 66.2 % (74.9 % on validation, against 77.0 % from scratch), and freezing the encoder during the first five epochs only narrows the gap (66.6 %). The learning curves of Figure 9 show that this is not an artefact of the large training set: the two initialisations are indistinguishable from 5 to 25 examples per sign (the largest difference, 1.4 points at 5 examples, is within one seed standard deviation) and the model trained from scratch is ahead from 50 examples per sign onwards (by 2.0 points with all data).

Is the ASL encoder informative at all? To rule out a poor fine-tuning recipe, we trained only a new classifier on top of the frozen ASL encoder (a linear probe). With all LSF training data it reaches 38.3 % validation top-1, far below the model trained from scratch (77.0 %) and even below the logistic regression on hand-crafted temporal statistics (57.0 %). The representation learnt on ASL thus retains some information about LSF signs, but much less than simple features of the target data: the absence of transfer is not due to the optimisation of the fine-tuning stage alone.

A gentler fine-tuning does not change the picture either: with 25 examples per sign, a five times lower learning rate gives 46.3 % validation top-1, below both the standard fine-tuning (50.6 %) and training from scratch (51.5 %) for the same seed. These two checks were run with a single seed on the laptop GPU.

6.4 Efficiency and deployment

Table 5 summarises the cost of inference. Preprocessing takes about 0.3 ms per clip in Python, and a sign is classified only once, when the segmenter detects its end; the Conv-Transformer then needs 7.8 ms natively and 16 ms in the browser, and the BiGRU 8 ms. Landmark extraction by MediaPipe, which runs on every frame, is by far the dominant cost of the live demo.

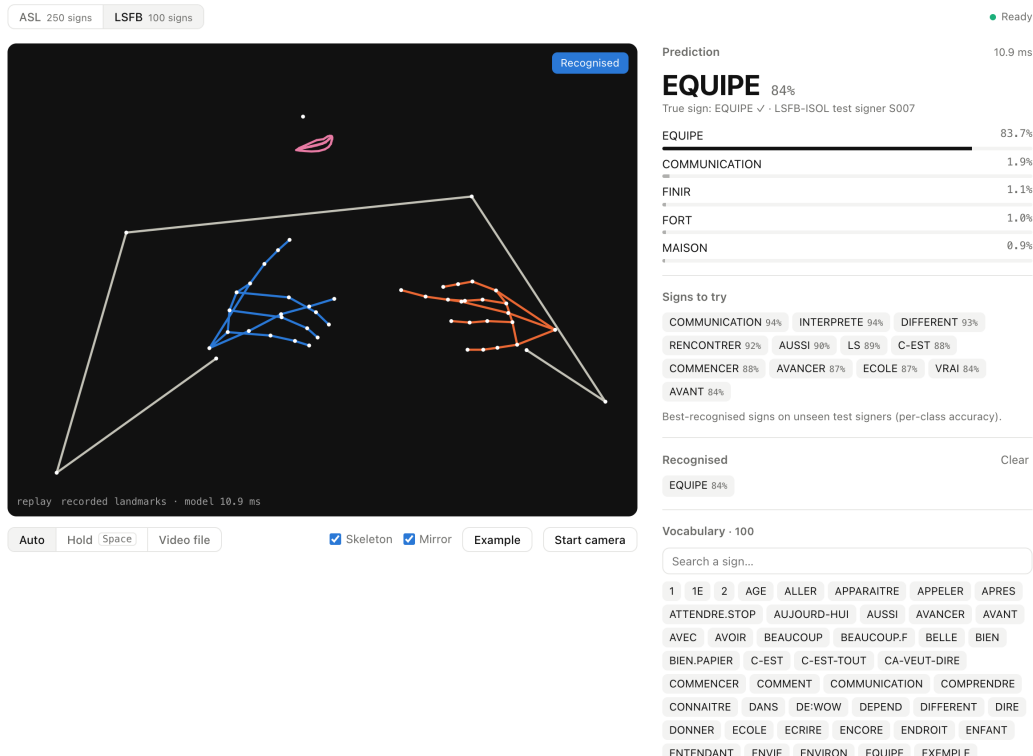


Figure 10. The browser demo replaying an LSFB-ISOL clip of an unseen test signer: skeleton, top-5 predictions with the true sign, best-recognised signs to try, history and vocabulary. The demo ships one test clip for each of 40 glosses (the model recognises 70.0 % of them); with a webcam, signs are segmented automatically from hand presence and motion.

Quantisation deserves a comment. Applied naively, dynamic INT8 quantisation of the Conv-Transformer reduces its validation top-1 from 68 % to 34 %. The cause is a small number of very large inputs: when a detected hand is tiny, the hand-local features (Section 4.1) divide by a near-zero length and reach values close to 200, while 99.9 % of them stay below 6.2. A per-tensor dynamic scale on the first layer then wipes out the resolution of all the other inputs. Keeping this single layer in FP32 restores the accuracy entirely. The quantised model is $3.3\times$ smaller, but it is $2.1\times$ slower on the CPU, natively and in WebAssembly, because quantising and dequantising activations around many small matrix products costs more than the integer arithmetic saves at this scale. Since the demo already downloads about 25 MB for MediaPipe, we ship the FP32 model.

7 Discussion

What the numbers mean. All results are obtained on people absent from training, which is the setting that matters for a deployed system and, as Section 6.2 shows, a much harder one than a random split suggests. The remaining errors have three main sources that call for different remedies: signs that are genuinely ambiguous in a closed vocabulary (near-synonyms, pointing and directional signs), landmark detection failures, which dominate the hardest clips, and variation between signers, which is larger than the differences between our two architectures. More signers and better hand tracking are likely to help more than a larger classifier.

Why does ASL pre-training not transfer? We expected an ASL encoder to help, at least when few LSFB examples are available, since both languages build signs from handshapes, movements and locations. The two corpora, however, differ in much more than the language. ASL clips are citation forms recorded one at a time on a smartphone; LSFB clips are cut out of spontaneous conversations filmed in a studio, are

less than half as long (median 0.32 s against 0.73 s) and are co-articulated with their neighbours. Hands are missing in a third of the ASL frames but in 1.5 % of the LSFB frames, and the landmarks come from two generations of MediaPipe. A pre-trained encoder is therefore tuned to a different distribution of inputs, and with even a few hundred labelled clips the target corpus provides more relevant information than this prior. We stress that this is a hypothesis compatible with our measurements, not a demonstrated cause: separating language from recording conditions would require a second ASL corpus of conversational signs or an LSFB corpus of citation forms. The practical lesson is that cross-lingual pre-training on pose sequences is not a free lunch; self-supervised pre-training on unlabelled, in-domain continuous signing (for example LSFB-CONT [11] or MEDI-API-SKEL [6] for French Sign Language) is a more promising direction.

Lessons for practitioners. Several of our findings are cheap to check and easy to miss. Image-space translation and scaling augmentations are cancelled by a body-centred normalisation and should be applied after it. Normalised landmark coordinates are anisotropic whenever pixels or frames are not square, which silently distorts the body frame across datasets. Early stopping on a plateau while the learning rate is still high can make an ablation look much worse than it is. Rare feature outliers that are harmless in floating point can break post-training quantisation, and quantisation does not necessarily make a small model faster on a CPU. Finally, a preprocessing function duplicated in two languages must be tested for parity, which costs little once fixtures are generated from the reference implementation.

Limitations. The study has several limitations. (i) The ASL test set contains four signers only, so the per-signer estimates are uncertain, and LSFB clips come from a single corpus recorded in one studio setting. (ii) Both vocabularies are small and fixed; out-of-vocabulary signs are always mapped to the closest known sign, and the demo has no rejection class. (iii) Our absolute ASL accuracy is well below that of the best competition solutions [8], which relied on heavier regularisation and augmentation, much longer schedules and ensembles of several models; a longer schedule alone did not help ours, and the ablations were run with one seed, 4 of them ending before the full schedule. (iv) The only non-manual information used is the lip contour; eyebrows, gaze and head movements, which carry grammatical information in sign languages, are ignored. (v) The demo segments signs from hand presence and motion, which works for signs produced one at a time with a pause but not for natural, continuous signing. (vi) No Deaf signer was involved in the evaluation of the demo; usefulness for the community remains to be established with its members.

Ethical considerations. We use two datasets released for research by their creators, under the Kaggle competition rules and the CC BY 4.0 licence (LSFB-ISOL), and we never redistribute ASL data; the demo only ships a few LSFB clips, with attribution. The browser demo processes video locally and stores nothing; only landmarks are computed, but face and body landmarks can still identify a person, so any extension that records them should obtain explicit consent. We describe the system as *isolated sign recognition*: presenting such a classifier as a translator would mislead users about its abilities and could be harmful in settings where communication matters. A model card summarising intended uses, out-of-scope uses and known biases is released with the code [21].

8 Conclusion

We built and evaluated an isolated sign recognition system that runs entirely in a web browser, with a signer-independent protocol throughout. A light convolution–Transformer on MediaPipe landmarks recognises 250 ASL signs from unseen signers with 64.8 ± 0.4 % top-1 accuracy, and 100 LSFB glosses with 68.3 %. Two findings deserve attention beyond the numbers. First, careful evaluation changes the conclusions: random splits inflate accuracy by 16.9 points, and an early-stopping artefact multiplied the apparent effect of one ablation by five. Second, cross-lingual pre-training from ASL did not help LSFB, even with five examples per sign, which suggests that differences in recording and elicitation

conditions outweigh what the two languages share at the level of pose features. Future work includes self-supervised pre-training on in-domain continuous signing, a rejection class for out-of-vocabulary signs, richer non-manual features, and an evaluation of the demo with Deaf signers.

References

- [1] S. Albanie, G. Varol, L. Momeni, T. Afouras, J. S. Chung, N. Fox, and A. Zisserman. BSL-1K: Scaling up co-articulated sign language recognition using mouthing cues. In *European Conference on Computer Vision (ECCV)*, 2020.
- [2] V. Bazarevsky, I. Grishchenko, K. Raveendran, T. Zhu, F. Zhang, and M. Grundmann. BlazePose: On-device real-time body pose tracking. *arXiv preprint arXiv:2006.10204*, 2020.
- [3] V. Belissen, A. Braffort, and M. Gouiffès. Dicta-Sign-LSF-v2: Remake of a continuous French Sign Language dialogue corpus and a first baseline for automatic sign language processing. In *Proceedings of the Twelfth Language Resources and Evaluation Conference (LREC)*, 2020.
- [4] M. Boháček and M. Hruží. Sign pose-based transformer for word-level sign language recognition. In *IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, 2022.
- [5] D. Bragg, O. Koller, M. Bellard, L. Berke, P. Boudreault, A. Braffort, N. Caselli, M. Huenerfauth, H. Kacorri, T. Verhoef, C. Vogler, and M. Ringel Morris. Sign language recognition, generation, and translation: An interdisciplinary perspective. In *Proceedings of the 21st International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS)*, 2019.
- [6] H. Bull, A. Braffort, and M. Gouiffès. MEDI-API-SKEL – a 2D-skeleton video database of French Sign Language with aligned French subtitles. In *Proceedings of the Twelfth Language Resources and Evaluation Conference (LREC)*, 2020.
- [7] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2014.
- [8] A. Chow, G. Cameron, M. Georg, M. Sherwood, P. Culliton, S. Sepah, S. Dane, and T. Starner. Google – isolated sign language recognition. Kaggle competition, <https://www.kaggle.com/competitions/asl-signs>, 2023.
- [9] Y. N. Dauphin, A. Fan, M. Auli, and D. Grangier. Language modeling with gated convolutional networks. In *International Conference on Machine Learning (ICML)*, 2017.
- [10] A. Desai, L. Berger, F. O. Minakov, V. Milan, C. Singh, K. Pumphrey, R. E. Ladner, H. Daumé III, A. X. Lu, N. Caselli, and D. Bragg. ASL Citizen: A community-sourced dataset for advancing isolated sign language recognition. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023.
- [11] J. Fink, B. Frénay, L. Meurant, and A. Cleve. LSFB-CONT and LSFB-ISOL: Two new datasets for vision-based sign language recognition. In *International Joint Conference on Neural Networks (IJCNN)*, 2021.
- [12] A. Gulati, J. Qin, C.-C. Chiu, N. Parmar, Y. Zhang, J. Yu, W. Han, S. Wang, Z. Zhang, Y. Wu, and R. Pang. Conformer: Convolution-augmented transformer for speech recognition. In *Interspeech*, 2020.

- [13] H. Hu, W. Zhao, W. Zhou, Y. Wang, and H. Li. SignBERT: Pre-training of hand-model-aware representation for sign language recognition. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.
- [14] S. Jiang, B. Sun, L. Wang, Y. Bai, K. Li, and Y. Fu. Skeleton aware multi-modal sign language recognition. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2021.
- [15] Y. Kartynnik, A. Ablavatski, I. Grishchenko, and M. Grundmann. Real-time facial surface geometry from monocular video on mobile GPUs. *arXiv preprint arXiv:1907.06724*, 2019.
- [16] O. Koller. Quantitative survey of the state of the art in sign language recognition. *arXiv preprint arXiv:2008.09918*, 2020.
- [17] D. Li, C. Rodriguez, X. Yu, and H. Li. Word-level deep sign language recognition from video: A new large-scale dataset and methods comparison. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2020.
- [18] I. Loshchilov and F. Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.
- [19] C. Lugaresi, J. Tang, H. Nash, C. McClanahan, E. Uboweja, M. Hays, F. Zhang, C.-L. Chang, M. G. Yong, J. Lee, W.-T. Chang, W. Hua, M. Georg, and M. Grundmann. MediaPipe: A framework for building perception pipelines. *arXiv preprint arXiv:1906.08172*, 2019.
- [20] L. Meurant. Corpus LSFB. un corpus informatisé en libre accès de vidéos et d’annotations de la langue des signes de Belgique francophone (LSFB). Laboratoire de Langue des Signes de Belgique Francophone (LSFB-Lab), University of Namur, <https://www.corpus-lsfb.be>, 2015.
- [21] M. Mitchell, S. Wu, A. Zaldivar, P. Barnes, L. Vasserman, B. Hutchinson, E. Spitzer, I. D. Raji, and T. Gebru. Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*, 2019.
- [22] ONNX Runtime developers. ONNX Runtime. <https://onnxruntime.ai>, 2021.
- [23] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna. Rethinking the inception architecture for computer vision. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [24] A. Tunga, S. V. Nuthalapati, and J. Wachs. Pose-based sign language recognition using GCN and BERT. In *IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, 2021.
- [25] H. R. Vaezi Joze and O. Koller. MS-ASL: A large-scale data set and benchmark for understanding American Sign Language. In *British Machine Vision Conference (BMVC)*, 2019.
- [26] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [27] S. Yan, Y. Xiong, and D. Lin. Spatial temporal graph convolutional networks for skeleton-based action recognition. In *AAAI Conference on Artificial Intelligence*, 2018.
- [28] K. Yin, A. Moryossef, J. Hochgesang, Y. Goldberg, and M. Alikhani. Including signed languages in natural language processing. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2021.
- [29] J. Yosinski, J. Clune, Y. Bengio, and H. Lipson. How transferable are features in deep neural networks? *Advances in Neural Information Processing Systems (NeurIPS)*, 2014.

- [30] F. Zhang, V. Bazarevsky, A. Vakunov, A. Tkachenka, G. Sung, C.-L. Chang, and M. Grundmann. MediaPipe Hands: On-device real-time hand tracking. *arXiv preprint arXiv:2006.10214*, 2020.

A Hyper-parameters

Table 6. Hyper-parameters shared by all experiments unless stated otherwise. All values are read from the configuration files released with the code.

Group	Parameter	Value
Preprocessing	frames T' / length mode	64 / linear resampling
	coordinates	(x, y) ; depth dropped
	landmark parts	both hands, upper body (7 points), lips (40 points)
	features per frame	positions, dominant-hand local shape, velocities ($F = 440$)
Augmentation	spatial (body frame), probability	0.8
	rotation / per-axis scale / shift	$\pm 15^\circ$ / $[0.8, 1.2]$ / ± 0.1 shoulder widths
	global speed / local warp	$[0.8, 1.2]$ / $\pm 30\%$ over 4 segments
	frame / landmark / hand dropping	0.1 / 0.05 / 0.05
BiGRU	Gaussian noise (image units)	0.002
BiGRU	input projection / hidden / layers	192 / 192 per direction / 2
	pooling / dropout	masked mean max / 0.3
Conv-Transformer	width / stages	192 / 2
	per stage	3 conv. blocks + 1 Transformer block
	kernel / expansion / heads / MLP ratio	17 / 2 / 4 / 2
	dropout	0.2
Optimisation	optimiser	AdamW, lr 10^{-3} , weight decay 0.05
	schedule	3 warm-up epochs, cosine decay
	batch / max. epochs / patience	128 / 50 / 10 (validation macro-F1)
	label smoothing / gradient clipping	0.1 / 1.0
	precision	mixed (fp16) on GPU

B Landmark subset

Table 7. Unified landmark subset, in the order used by the model (MediaPipe indices).

Part	Points	MediaPipe indices
Left hand	21	hand model 0–20
Right hand	21	hand model 0–20
Upper body	7	pose 0 (nose), 11–12 (shoulders), 13–14 (elbows), 15–16 (wrists)
Lips	40	face mesh: outer contour (20) and inner contour (20)
Total	89	